DRHS DS-TEAM2 · 2026 VIRGINIA DATATHON · ENERGY

# Who Pays for the Cloud?

**The Cloud Dividend:** using Virginia's new data-center tax to protect the households least able to afford rising power bills

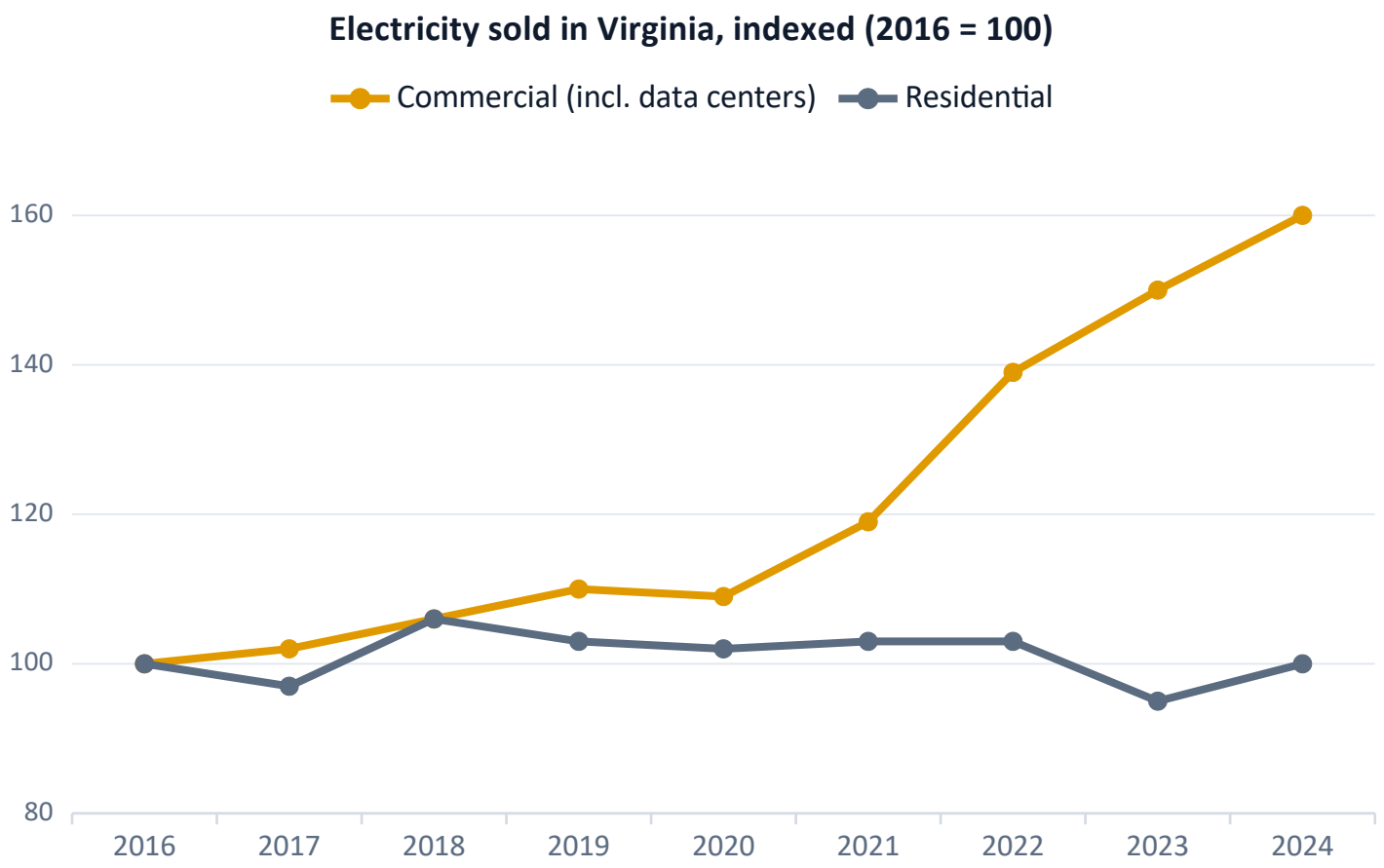
**$122M**

a year for energy relief, from 20% of the data-center electricity tax

Rayhan · James · Ibrahim · Dyson

THE TREND

# Data centers are pushing everyone's power costs up




**+60%**

commercial electricity sales since 2016, while homes stayed flat

**+26%**

Virginia residential price per kWh, 2022 → 2026

**+$14–37**

per month on a typical Dominion bill by 2040 from data-center growth alone (JLARC)

Sources: EIA retail sales via data.virginia.gov (Datathon 2026); EIA Electric Power Monthly 5.6.B (2026 = Jan–Jul YTD); JLARC Rpt. 598 (2024), Table 4-2.

WHO CAN'T ABSORB IT

# Low-income homes spend 7× more of their income on energy

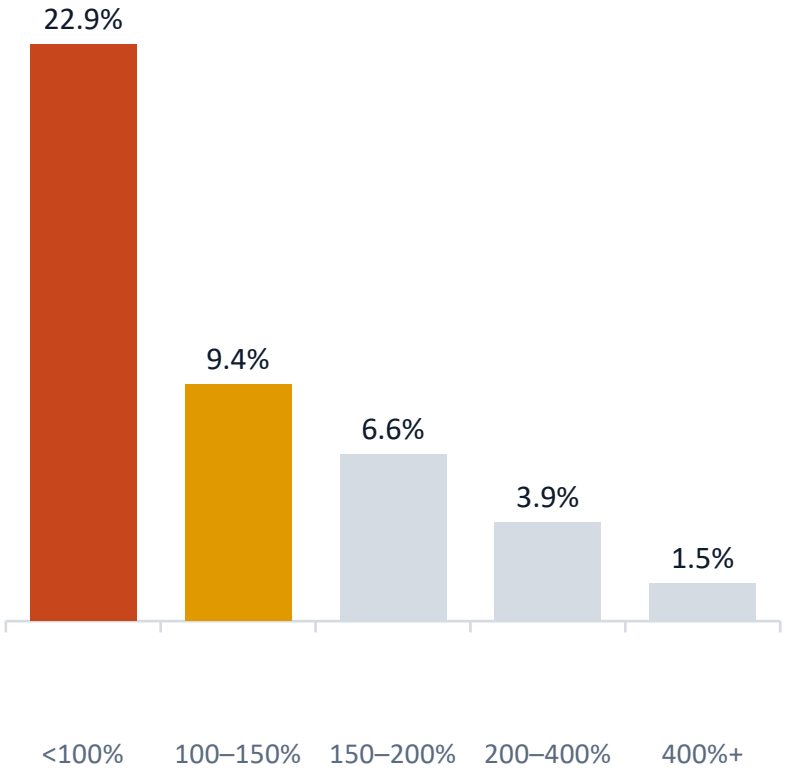
**14.3%**

of income on home energy for households at or below 150% of poverty

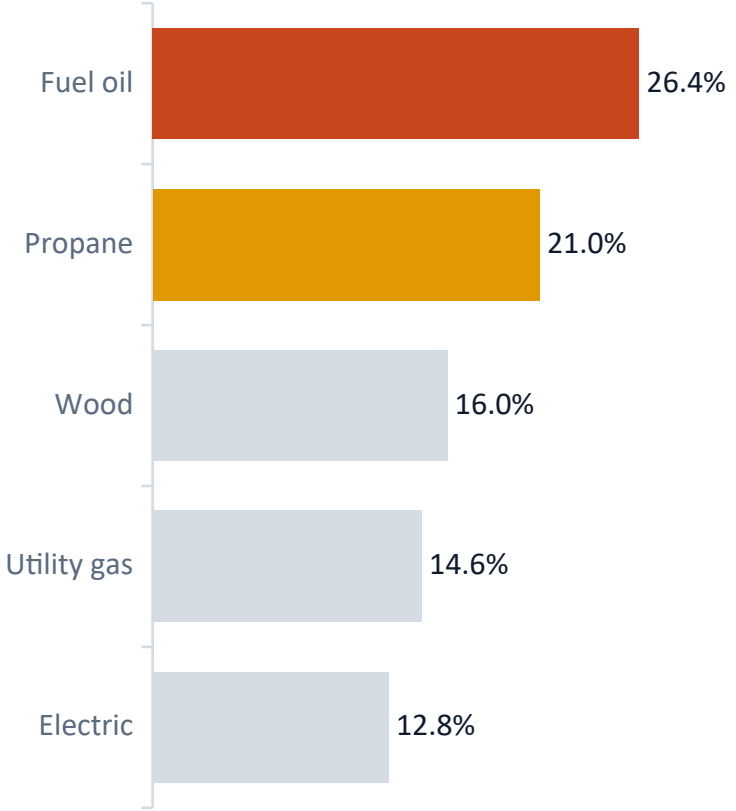
**2.1%**

for the average Virginia household. 6% is the standard for “affordable.”

**Burden by income (% of poverty line)**



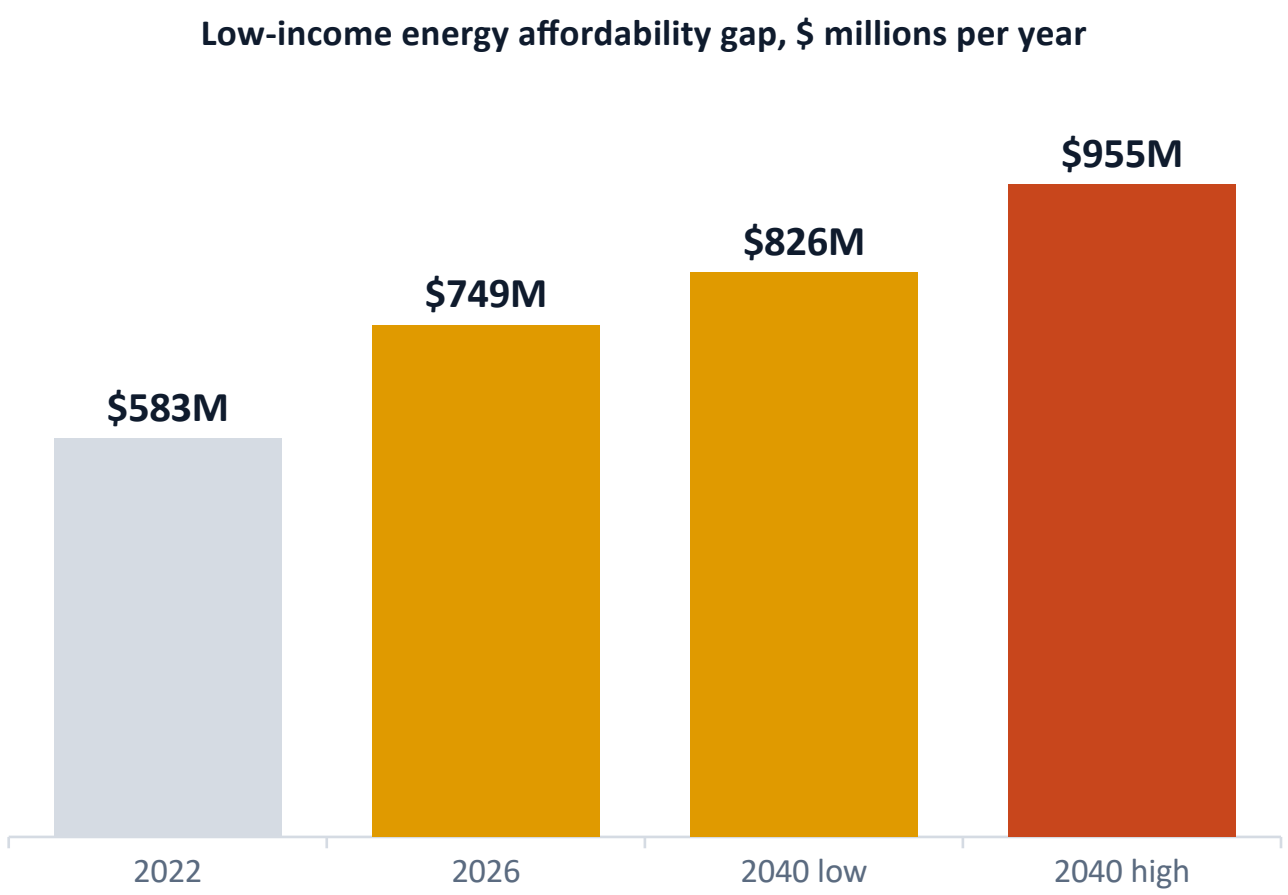

**Low-income burden by heating fuel**




Source: DOE LEAD Tool 2022 (ACS 2018–22 microdata calibrated to utility bills), updated to 2026 EIA prices; 466,049 households ≤150% of poverty, all 133 localities.

THE GAP

# Closing the gap costs $749M a year, and existing aid covers 11%




**$82M** paid by the Energy Assistance Program in 2025, about 11% of the gap

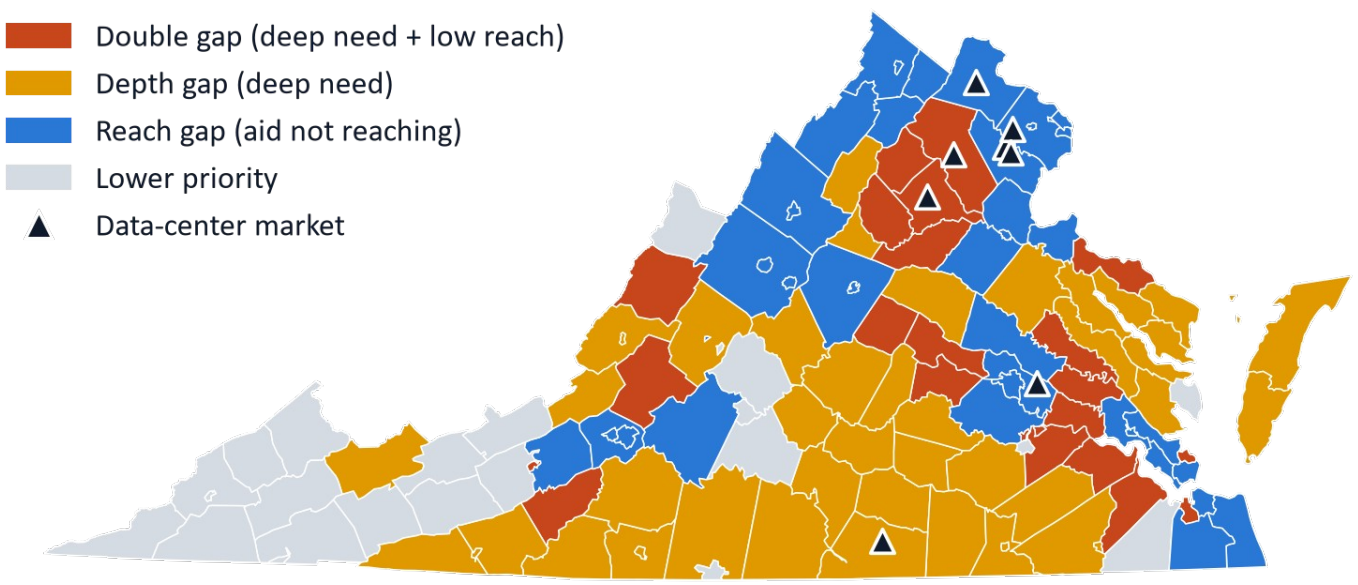
**28%** of eligible households reached; only 14% in data-center localities

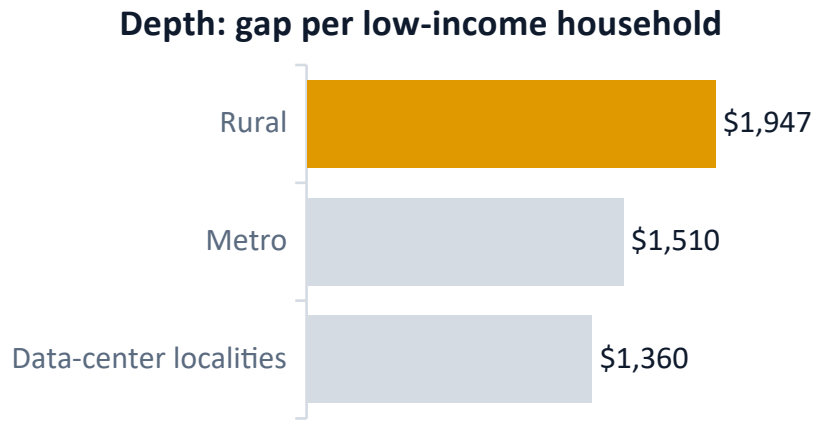
**276,934** SNAP households, already income-verified by local DSS, got no energy aid

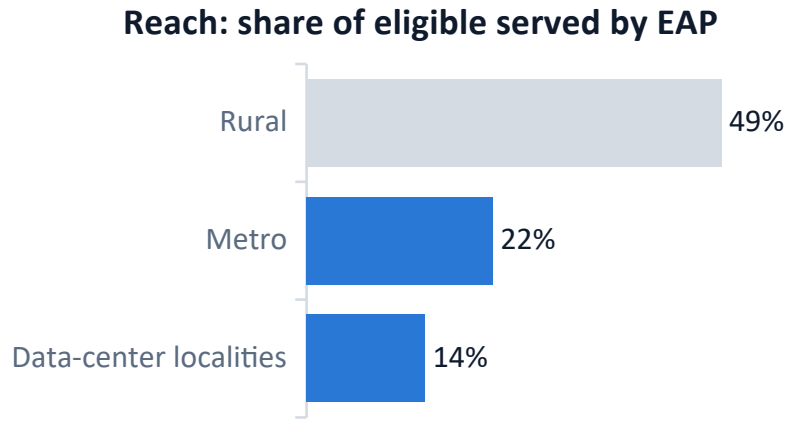
Sources: LEAD + EIA (gap = cost of bringing every household ≤150% FPL to 6% of income); JLARC 2040 scenarios; VDSS EAP FFY2025 & SNAP July 2026 (data.virginia.gov).

THE PATTERN

# Two different problems need two different fixes



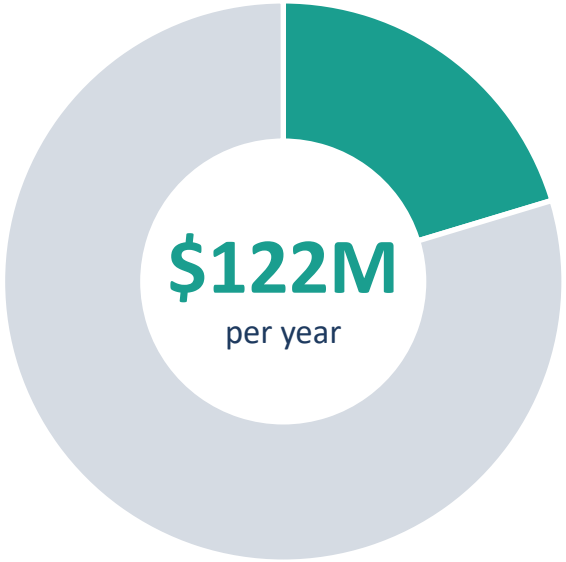




Sources: LEAD; VDSS EAP FFY2025; USDA ERS metro codes; data-center markets per JLARC 2024. Split at statewide medians of gap per household and EAP reach.

OUR SOLUTION

# The Cloud Dividend: the industry driving costs helps pay




20% of the ~$600M/yr data-center electricity tax → an Energy Relief Fund

LEVER 1 · REACH

## Auto-enroll

VDSS matches SNAP rolls to utility customer files and enrolls households in EAP and PIPP (bills capped at 6% of income). No new application. New York already does this.

**$58M/yr**

serves half of 276,934 SNAP households at the $415 avg. benefit

LEVER 2 · DEPTH

## Weatherize + heat pumps

Upgrade 40,021 low-income oil, propane and mobile homes in priority localities, 1/5 per year, building on HB2 (2026). Pays for itself in about 6 years.

**$64M/yr**

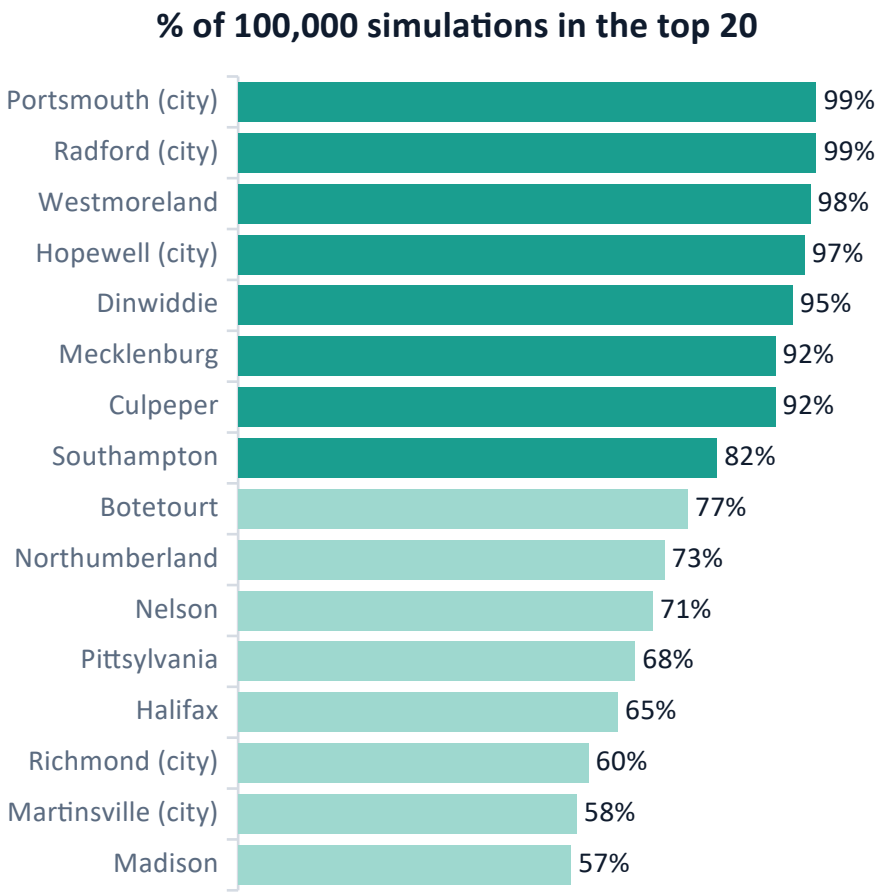
for 5 years at ~$8,000 per home; lowers bills permanently

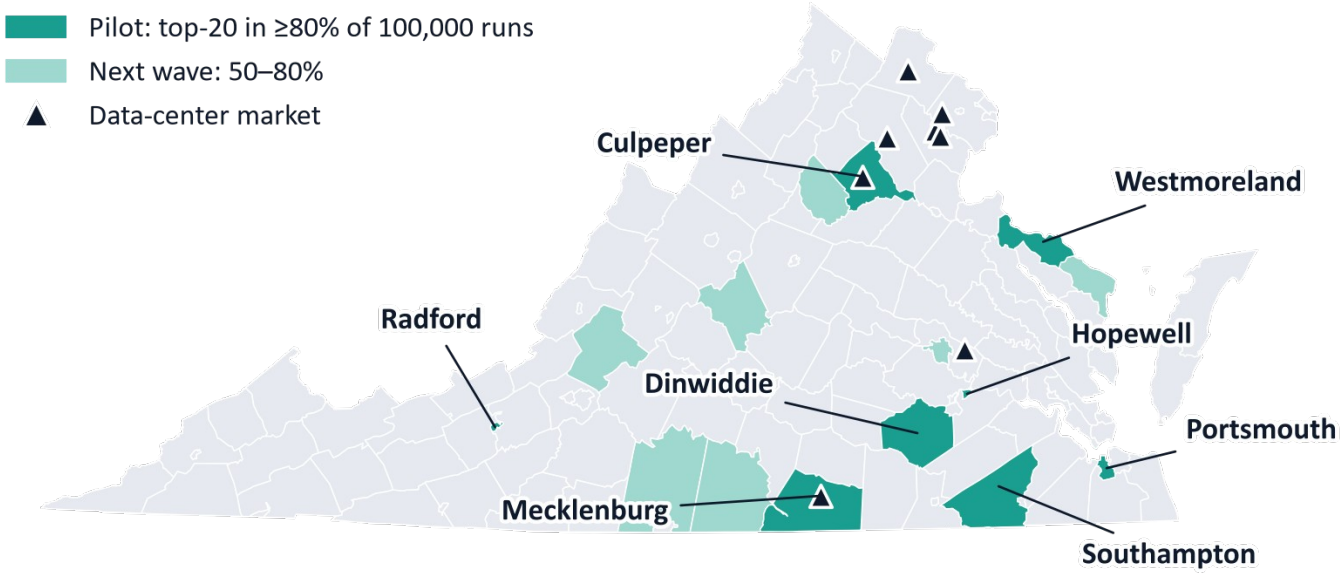
**Lever 3 · Target it:** an open Energy Burden Priority Index, refreshed yearly on data.virginia.gov

Sources: 2026 Appropriation Act (data-center electricity tax, $0.011/kWh, ~$600M/yr); VDSS PIPP; HB2 (2026); NY utility data-match law; state WAP cost plans; ACEEE.

WHERE TO START

# Eight localities stay top-20 in 80%+ of 100,000 simulations




## ~$10M/yr pilot in these 8 localities

24,017 low-income households · 16,477 SNAP households to auto-enroll · under 2% of the tax

Monte Carlo: 100,000 draws varying the 2040 price scenario, income growth, eligibility and index weights (severity, depth, scale, reach gap). Bars = % of draws in the top 20.

FEASIBILITY

# Every piece already exists, so Virginia can start in 2027

**1 2027 session**

Earmark 20% of the tax; extend it past its 2028 sunset

**2 Data-sharing**

VDSS–Dominion/APCo customer match (NY model)

**3 FY2028 pilot**

~$10M in the 8 priority localities

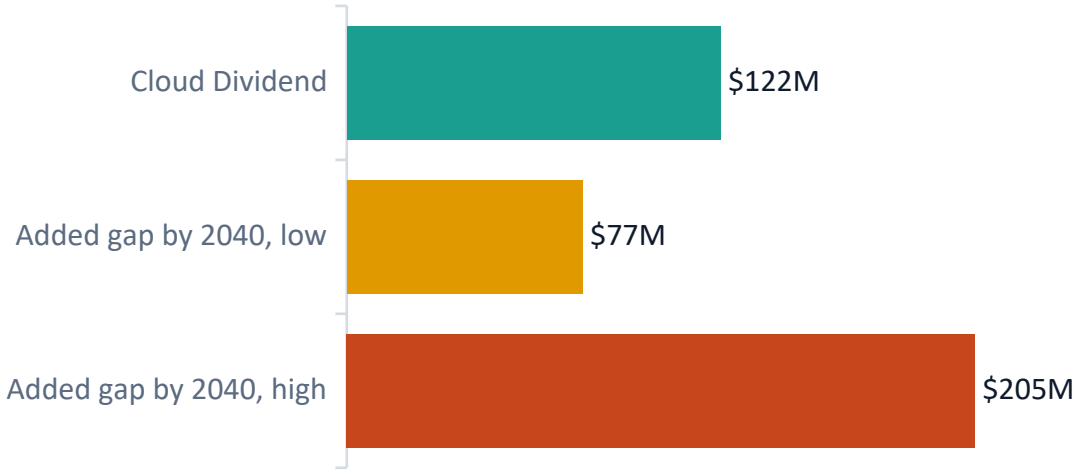
**4 FY2029 scale**

Statewide; track results with the open index

## ALREADY IN PLACE

- **Funding:** Data-center electricity tax, in effect since July 1, 2026
- **Delivery:** VDSS already runs EAP + PIPP in every locality
- **Mandate:** HB2 (2026) requires utilities to upgrade oil/propane homes
- **Targeting:** Our open index, built only on public data

**Covers all of the low case, ~60% of the high**




Data-center tax effective 7/1/2026 (sunsets 6/30/2028); EAP & PIPP administered by VDSS through local DSS offices; HB2 (2026); RGGI rejoined 7/1/2026 (DHCD efficiency funds).

# The industry driving costs up helps cover them

Start with a ~$10M pilot in 8 localities, then scale statewide with an open, yearly-updated priority index.

**$122M/yr**

Cloud Dividend for energy relief

**276,934**

SNAP households to auto-enroll

**40,021**

homes to weatherize or electrify

**Thank you. Questions?**